

Problem

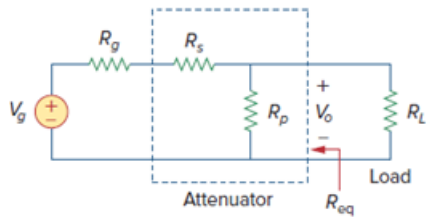
An attenuator is an interface circuit that reduces the voltage level without changing the output resistance.

(a) By specifying R_s and R_p of the interface circuit in Fig. 4.150, design an attenuator that will meet the following requirements:

$$\frac{V_o}{V_g} = 0.125, \quad R_{eq} = R_{Th} = R_g = 100 \, \Omega$$

(b) Using the interface designed in part (a), calculate the current through a load of $R_L = 50 \, \Omega$ when $V_g = 12 \, \text{V}$.

Figure. 4.150



Step-by-step solution

Step 1 of 5

$$(A) \quad \frac{V_o}{V_g} = \frac{R_p}{(R_g + R_s + R_p)} \rightarrow (1)$$

$$R_{eq} = R_p \parallel (R_g + R_s) = R_g.$$

$$R_g = \frac{R_p(R_g + R_s)}{(R_p + R_g + R_s)}.$$

$$R_g R_p + R_g^2 + R_g R_s = R_p R_g + R_p R_s.$$

$$R_p R_s = R_g(R_g + R_s) \rightarrow (2)$$

Step 2 of 5

From (1),

$$\frac{R_p}{\alpha} = R_g + R_s + R_p.$$

$$R_g + R_s = R_p \left(\left(\frac{1}{\alpha} \right) - 1 \right) = R_p \frac{(1-\alpha)}{\alpha} \rightarrow (1a)$$

Combining (2) and (1a) gives,

$$R_s = \left[\frac{(1-\alpha)}{\alpha} \right] R_{eq} \rightarrow (3)$$

$$= \frac{(1-0.125)(100)}{0.125} = 700 \, \text{Ohms}.$$

Step 3 of 5

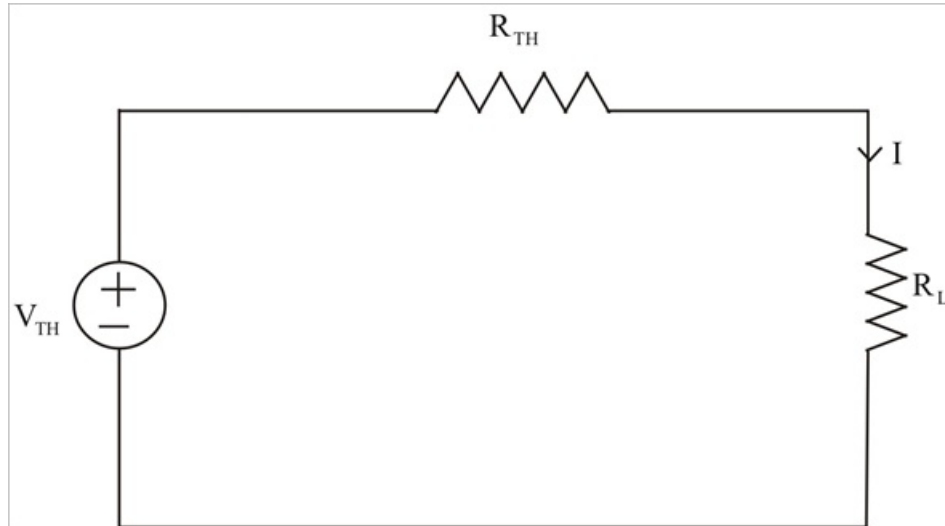
From (3) and (1a),

$$R_p \frac{(1-\alpha)}{\alpha} = R_g + \left[\frac{(1-\alpha)}{\alpha} \right] R_g = \frac{R_g}{\alpha} .$$

$$R_p = \frac{R_g}{(1-\alpha)} = \frac{100}{(1-0.125)} = 114.29 \text{ Ohms} .$$

Step 4 of 5

(B)



Step 5 of 5

$$V_{TH} = V_S = 0.125 V_g = 1.5 V .$$

$$R_{TH} = R_g = 100 \text{ Ohms} .$$

$$I = \frac{V_{TH}}{(R_{TH} + R_L)} = \frac{1.5}{150} = 10 \text{ mA} .$$